

Can a weaker middle signal soften the two-peak trough?

Triplet extension to the v5.2.2 study

65 / 75–80 / 90 MeV region — 10 September 2026

Two separate questions

This extension asks whether adding a weak middle component changes the predicted upper-limit trough, and whether the observed count spectra prefer such a component. The first question is tested by injecting a middle signal while retaining the parent continuum and outer peaks exactly. The second uses nested two- and three-component fits on identical fixed GP experiments. Agreement between a fitted limit curve and data is not treated as independent particle evidence.

Findings

A weak middle component can lift the single-signal limit response. It can soften a deficit-like middle response in 2015 or 2016, but the parent pair already predicts a positive middle response in 2021 and in the shared combinations. Adding another positive component there is an uplift of an existing central excess, rather than filling an initially negative trough.

The shared 2016+2021 fit chooses zero middle strength in its initial 75–80 MeV reference window. Matching the all-three fit’s shifted 2016 window, 78–83 MeV, instead gives a small weak-triplet gain of $2\Delta \log L = 1.109$ at 83 MeV. The all-three shared weak triplet gains 0.899 at its reference upper boundary. Both selected shared solutions reach a mass-domain edge. Larger gains with independent campaign rates can rearrange the outer strengths and no longer describe the same shared weak-triplet hypothesis.

Reference	Search	r_{pair}	r_{triplet}	$U_{\text{triplet}}/U_{\text{pair}}$
All-three fit	2015	-1.62	-1.57	1.015
All-three fit	2016	-0.72	-0.43	1.111
All-three fit	2021 (10%)	3.77	4.27	1.100
All-three fit	All three	2.75	3.34	1.145
16+21 fit	2016	-0.73	-0.44	1.107
16+21 fit	2021 (10%)	3.75	4.24	1.097
16+21 fit	2016 + 2021	2.87	3.43	1.135

The benchmark middle component has one-quarter of the weaker outer peak’s ϵ^2 and sits at the parent pair midpoint. Values are evaluated at the nearest half-MeV test coordinate. The signed profile root r retains deficits; positive values are excess-like within the conditional likelihood. Neither these roots nor fitted-mean curves are newly calibrated significance measures.

The study remains deterministic: no new toys are added. The native 2021 sample remains 10%, and the exploratory 2015 range still ends at 100 MeV. The previous v5.2.2 PDF is preserved alongside this extension.

1. Strength convention and the comparison that isolates the middle signal

The parent lower pair has shared separation 24.753 MeV in the all-three fit and 24.604 MeV in the 2016+2021 fit. The corresponding reference midpoints are 76.709 and 79.723 MeV. In the all-three construction the reference is 2015, with shifts $\delta_{2016} = +3.000$ and $\delta_{2021} = +2.514$ MeV. In the two-campaign construction the reference is 2016, with $\delta_{2021} = -0.509$ MeV. These observed-selected shifts remain fixed; they are not independently measured calibration corrections.

For $A = \epsilon^2/10^{-8}$, the middle injection is

$$A_{d,M} = f \min(A_{d,L}, A_{d,H}), \quad \bar{n}_d^{(3)} = \bar{n}_d^{(2)} + A_{d,M} S_d(u_M + \delta_d).$$

The full-support template S_d retains its mass-dependent count conversion and resolution. The original pair mean, continuum and outer amplitudes do not change. Fractions are $f = 0, 0.1, 0.25, 0.5, 1$. A fraction 0.25 in ϵ^2 corresponds to a factor 0.5 in the coupling ϵ . A smaller coupling need not yield fewer events than both outer components; the count conversion depends on mass and campaign.

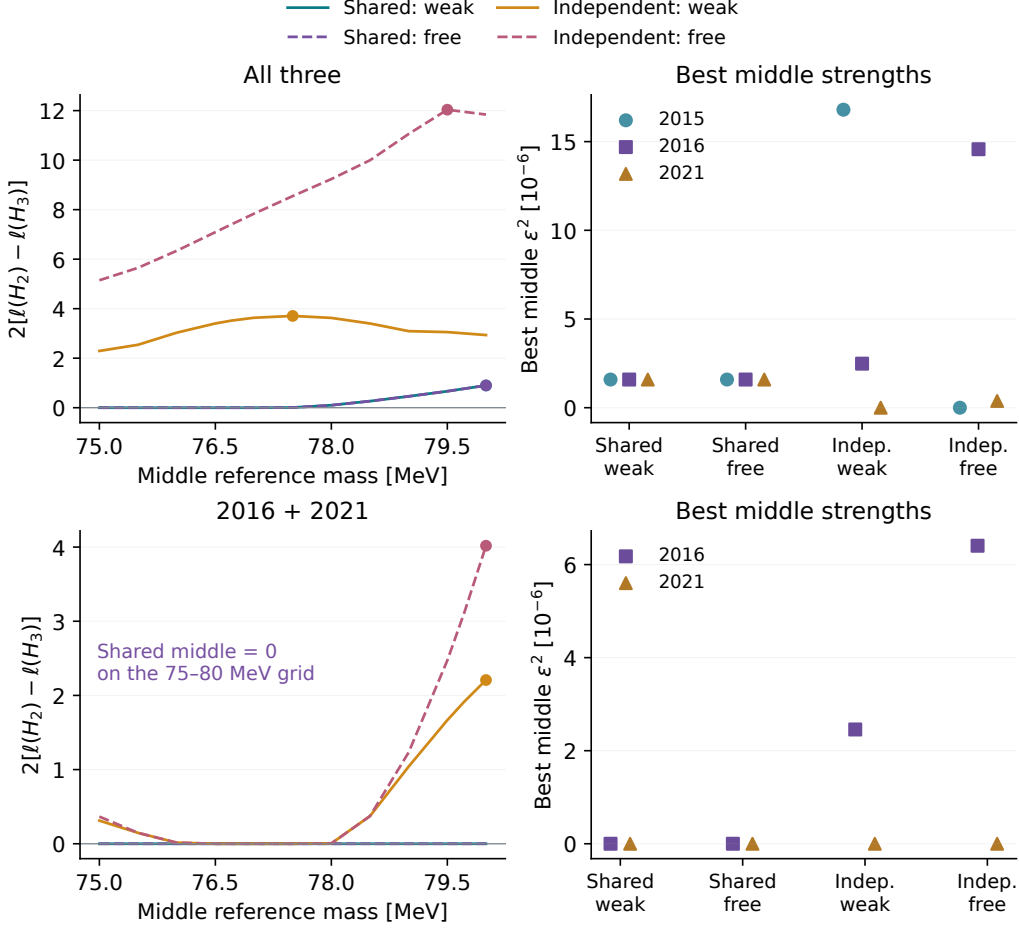
The middle reference coordinate runs from 75 to 80 MeV in 1 MeV steps, plus the exact parent midpoint. Its reconstructed mass in each campaign adds the parent offset. Thus the all-three reference endpoint 80 MeV means 80.83 and 82.514 MeV in 2015, 2016 and 2021. The shifted scan is explicitly broader than an unshifted 75–80 MeV interval. Two controls retain the midpoint’s middle event yield when moving its mass to 75 or 80 MeV, separating conversion changes from mass changes.

At every 0.5 MeV test coordinate, the moving GP prediction and covariance are recomputed from the complete mean spectrum using the inherited fixed/interpolated kernel states. Each year is tested at its shifted mass before stacking a shared-coupling likelihood with independent background nuisance blocks. The bounded 90% CL_s limit is solved with profiled observed and background-Asimov likelihood ratios, including the deficit branch [1,2].

The frozen-training control uses the GP mean and covariance obtained from the parent *pair* spectrum, then fits the same injected triplet counts. This isolates the middle component’s extra effect on GP training. Its background nuisance parameters remain profiled. At $f = 0$, the two controls coincide with the pair response exactly. Continuous mean spectra are deterministic inputs, not an assertion about an ensemble median.

Metrics use the nearest half-MeV point to the injected middle mass, an explicitly fixed ± 1 MeV interval, and a resolution interval of $\pm \sigma$ (the widest constituent resolution for combinations). The parent’s central-third bridge and full/outside-peak windows retain their original definitions. Log-limit distances use correlated mass points and have no chi-square or probability interpretation.

2. Do the count spectra prefer a weak third component?



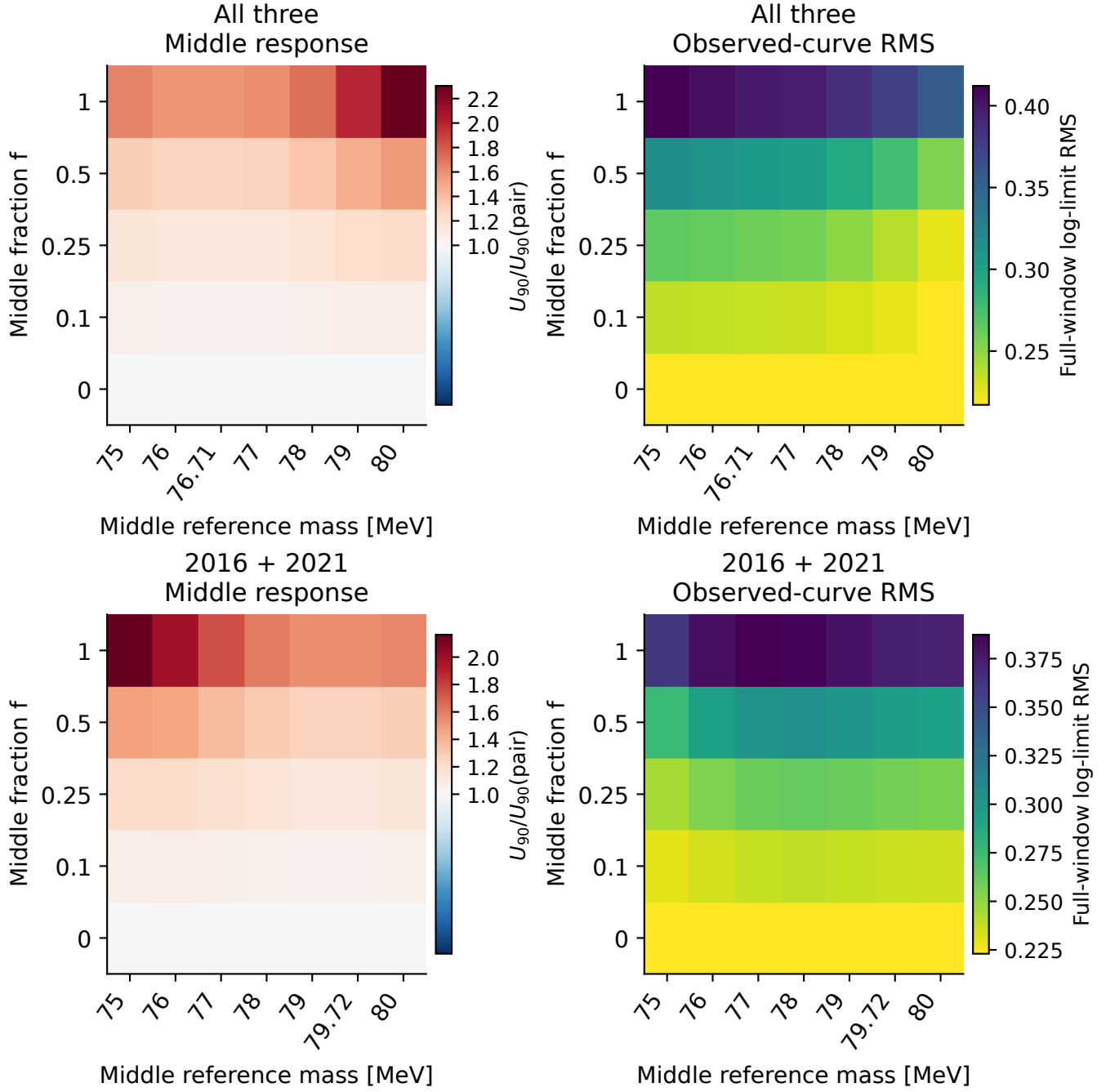
Two- versus three-component count fits keep the outer masses, offsets, fitted bins, GP prediction and covariance identical. Shared-rate fits use one amplitude per component across campaigns; independent-rate fits release that assumption. The displayed scan improvements are selected-model diagnostics, not fixed-mass or global p-values.

Case	Rate model	Weak middle gain	Unrestricted gain
All three	shared	0.899	0.899
All three	independent	3.707	12.037
16+21	shared	0.000	0.000
16+21	independent	2.207	4.018

The weak fit imposes $A_M \leq 0.5 \min(A_L, A_H)$ through the exact nonnegative coordinates $A_L = x_L + 2t$, $A_M = t$, $A_H = x_H + 2t$. Setting $t = 0$ recovers the pair. The unrestricted fit only imposes nonnegative amplitudes. Both scan middle reference mass 75–80 MeV in 0.5 MeV steps, plus the exact parent midpoint. All outer amplitudes may refit in this comparison, whereas the injection study keeps them fixed.

For shared 2016+2021, every scanned mass ties at zero middle strength; its stored first mass is not a measured location. The shared all-three fit reaches the 80 MeV upper boundary, with amplitudes $(A_L, A_M, A_H) = (382.72, 158.81, 374.85)$; its weaker-middle cap is inactive. Independent weak fits select zero middle strength in 2021, and some other campaigns reach the weak cap. These features limit a common-triplet interpretation.

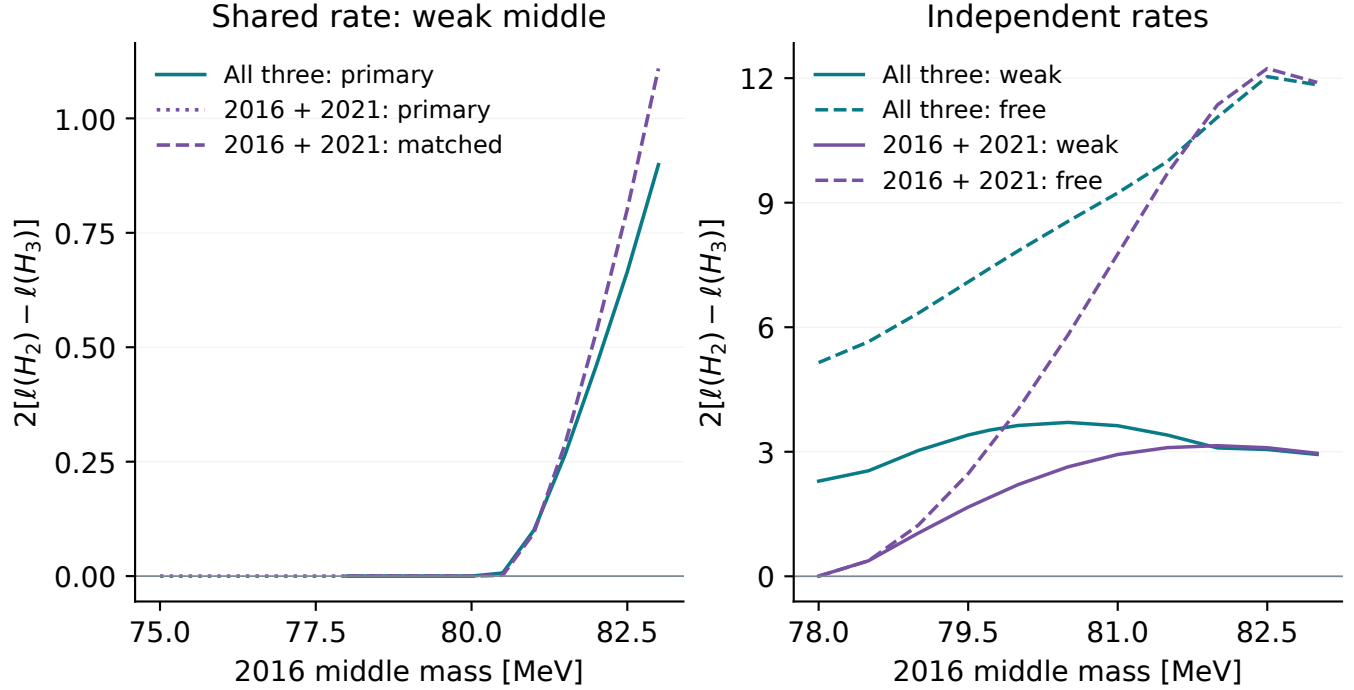
3. Where does a middle component raise the predicted limit?



Catalogue of reference middle locations and fractions of the weaker outer ϵ^2 . Ratios above one raise the limit relative to the parent pair; only a negative initial signed root can be described as a deficit being filled. Shape-distance panels, if shown, compare the full declared window with data and do not supply a fit probability.

4. A control that matches the shifted 2016 mass window

The original reference grids have different reconstructed meanings. The all-three 75–80 MeV reference interval maps to 78–83 MeV in 2016, whereas the original two-campaign grid uses 75–80 MeV directly in 2016. A separate control therefore scans 78–83 MeV for 2016+2021 on its unchanged count region, GP constraint, outer masses and original relative shift. Its 2021 offset differs from the all-three relative offset by 0.023 MeV; the original alignments are preserved rather than silently equated.

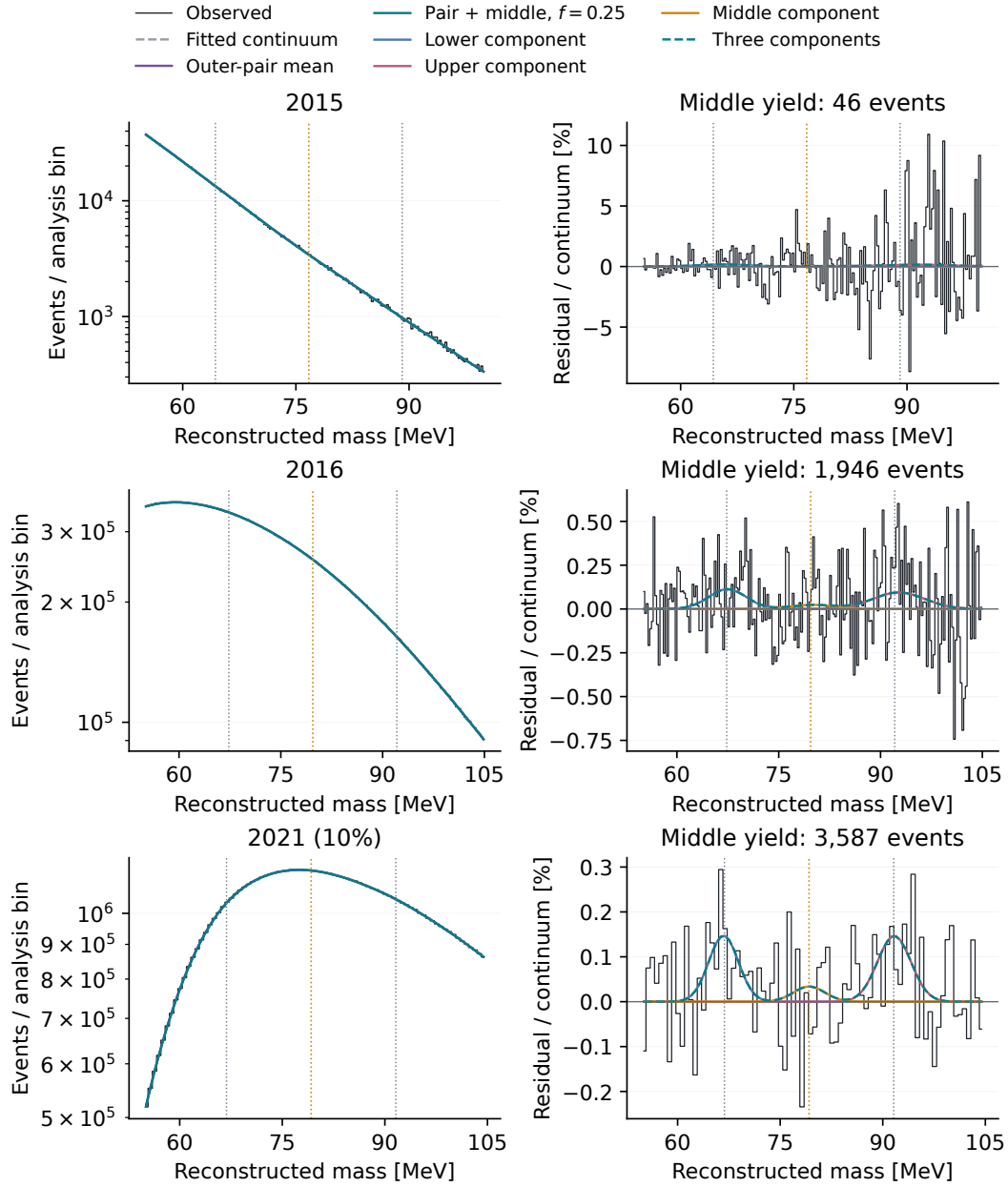


The matched 2016 window restores access to middle components near 82–83 MeV. Primary 75–80 MeV results are retained separately. Mass-selected gains remain exploratory and have no supplied trial correction.

The shared weak and unrestricted solutions coincide at 83 MeV, with $2\Delta \log L = 1.109$ and $(A_L, A_M, A_H) = (363.11, 177.37, 372.34)$. The middle fraction is 0.4885 of the weaker outer amplitude, just below the weak cap; the mass reaches the upper boundary. Independent weak and unrestricted fits gain 3.146 and 12.233, respectively. The much larger unrestricted gains can replace an outer component with the middle feature, so they do not demonstrate a small extra signal between two unchanged peaks.

This control changes the interpretation of the zero-middle primary result: it is a statement about that mass window, not evidence against every shifted triplet. The shared fits show a small preference at a scanned boundary; they do not locate a well-determined third mass or establish a common-particle explanation.

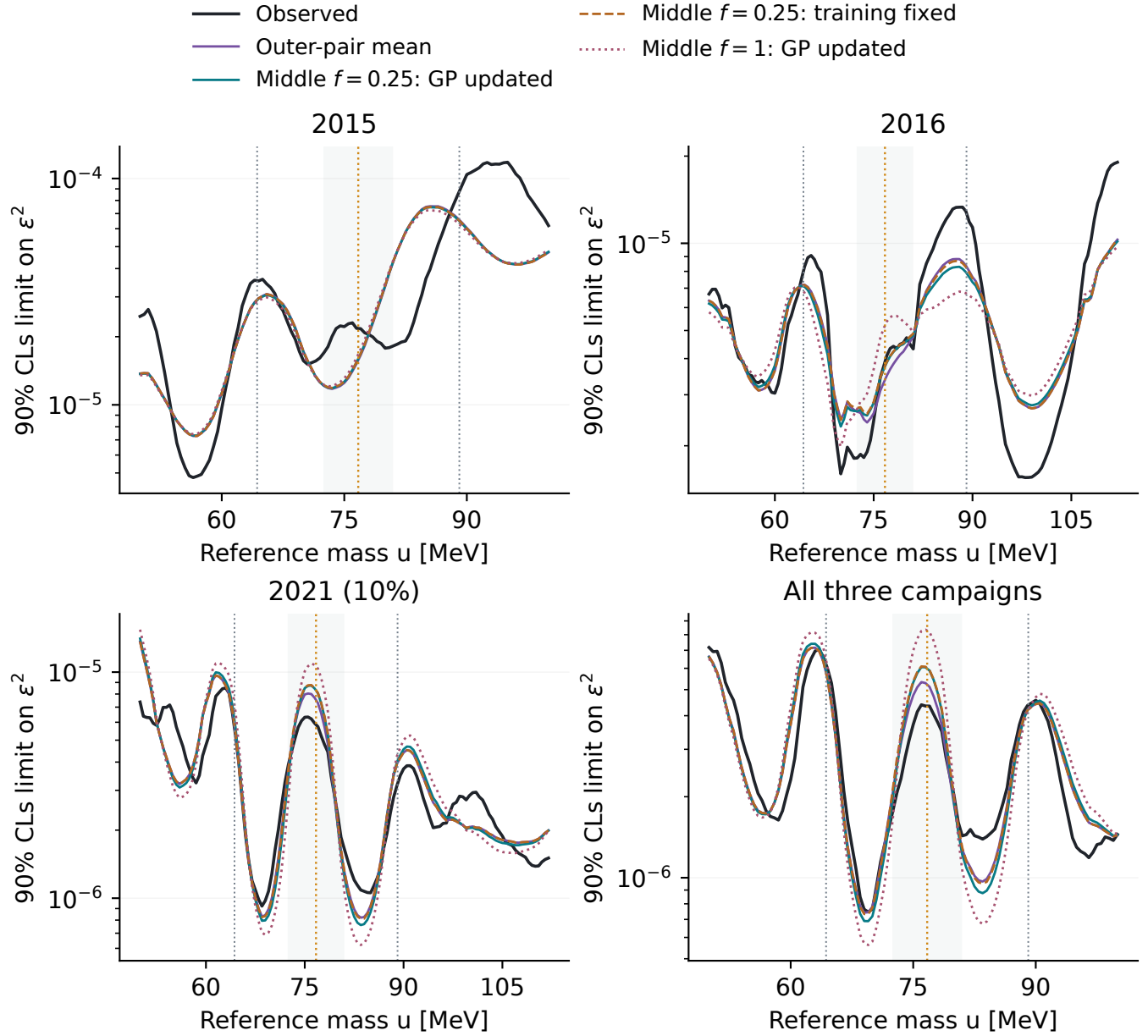
5. Lower triplet: all three campaigns – spectra and component sizes



Complete count-spectrum construction. The benchmark middle signal has $f = 0.25$ at the parent midpoint, with unchanged outer peaks and continuum. All tails are retained. Curve labels distinguish injected benchmarks from refitted means; these objects answer different questions.

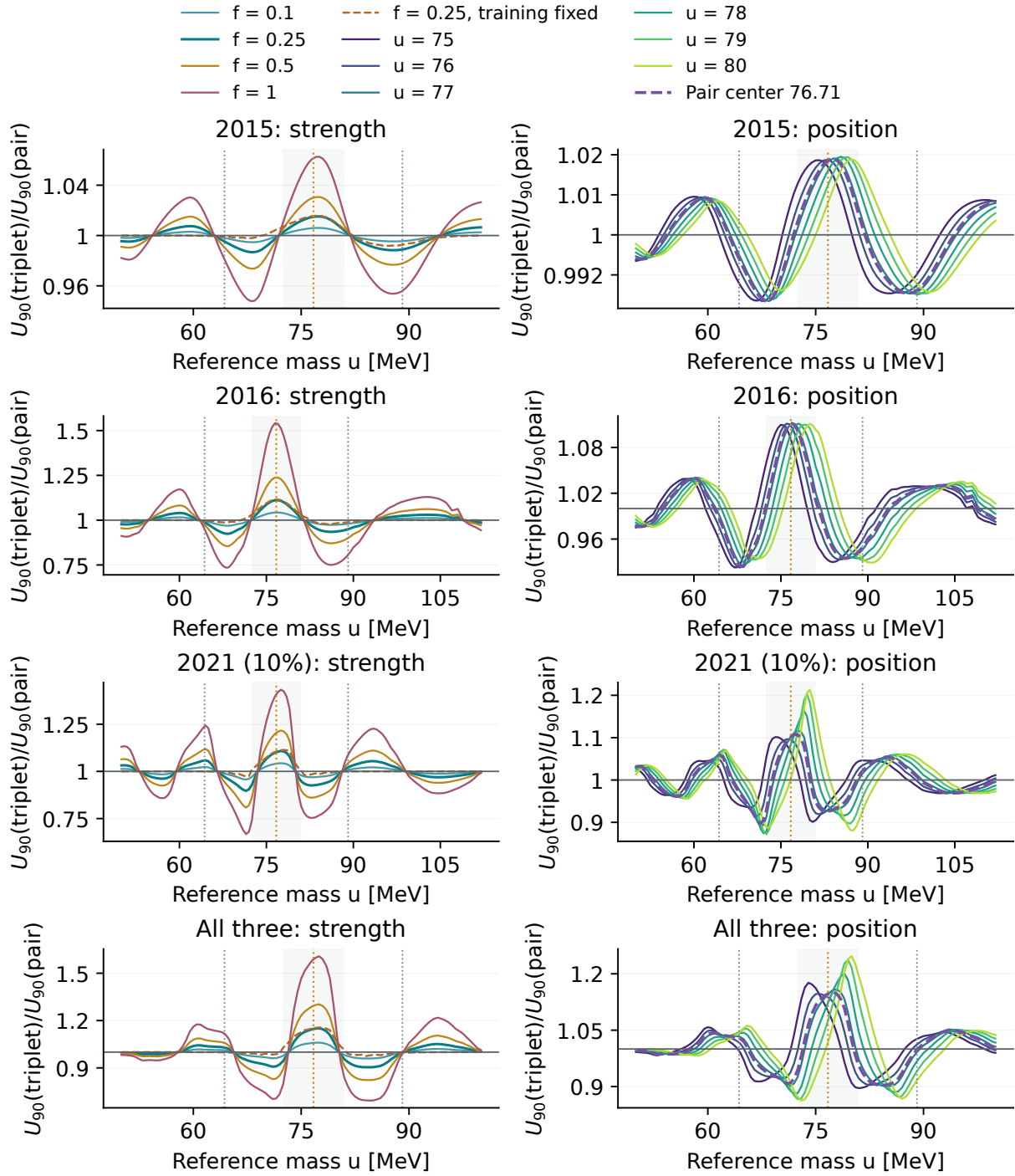
Campaign	m_M [MeV]	A_M	N_L	N_M	N_H
2015	76.71	79.69	724	46	62
2016	79.71	79.69	10449	1946	5802
2021	79.22	79.69	13111	3587	15071

6. Lower triplet: all three campaigns – upper-limit comparison



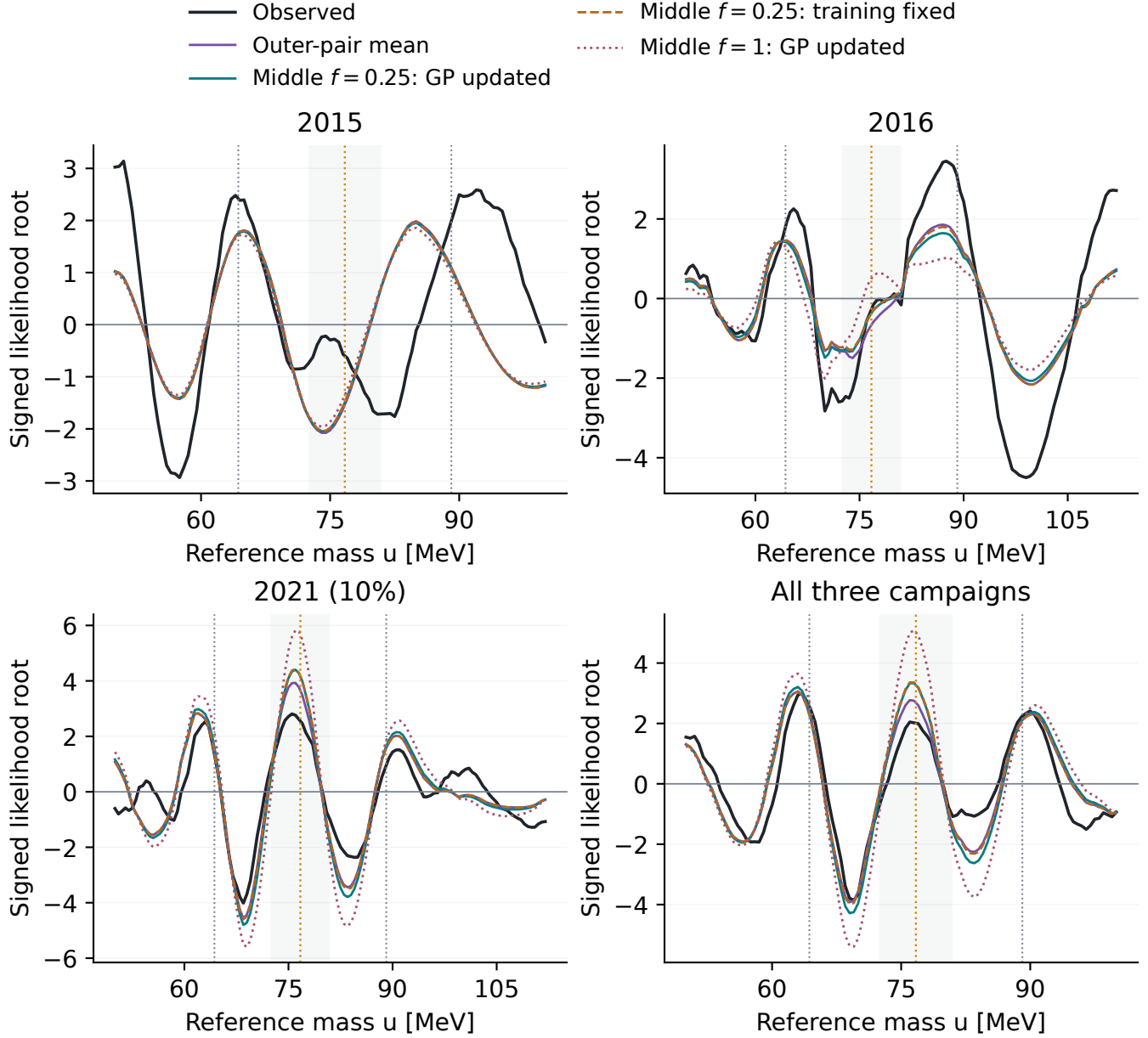
Actual 0.5 MeV bounded CL_s solutions for the observed spectrum, original pair and selected triplet constructions. The horizontal coordinate is the reference mass; each campaign uses its own shifted reconstructed test mass. Full fitted-mean controls refit the continuum and outer amplitudes, so their difference from the pair is not solely the added middle signal.

7. Lower triplet: all three campaigns – strength and middle-mass response



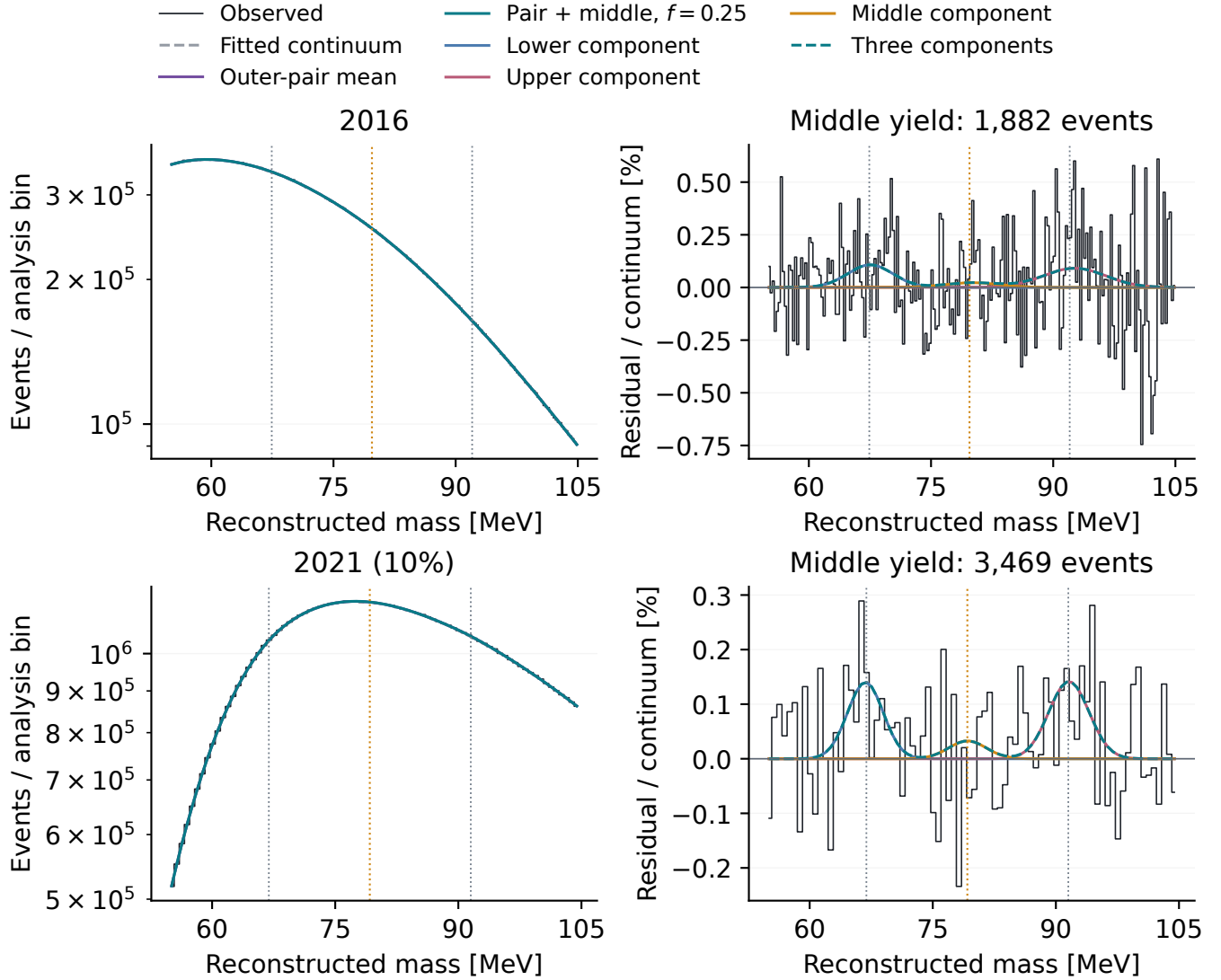
Middle-component propagation relative to the matched parent pair. Strength scans hold its location at the parent midpoint; location scans retain the displayed coupling fraction. Full and frozen-pair-training controls isolate extra sideband response. The $f = 1$ benchmark is outside the weak-fit cap $f \leq 0.5$.

8. Lower triplet: all three campaigns – which trough is being filled?



The signed root exposes the initial sign of the response. In 2021 and the shared combinations, the parent pair already generates a positive central response. Adding a middle component can strengthen that predicted excess. Positive-signal p_0 would plateau at 0.5 for deficits and obscure this distinction; root curves retain it without implying a calibrated triplet significance.

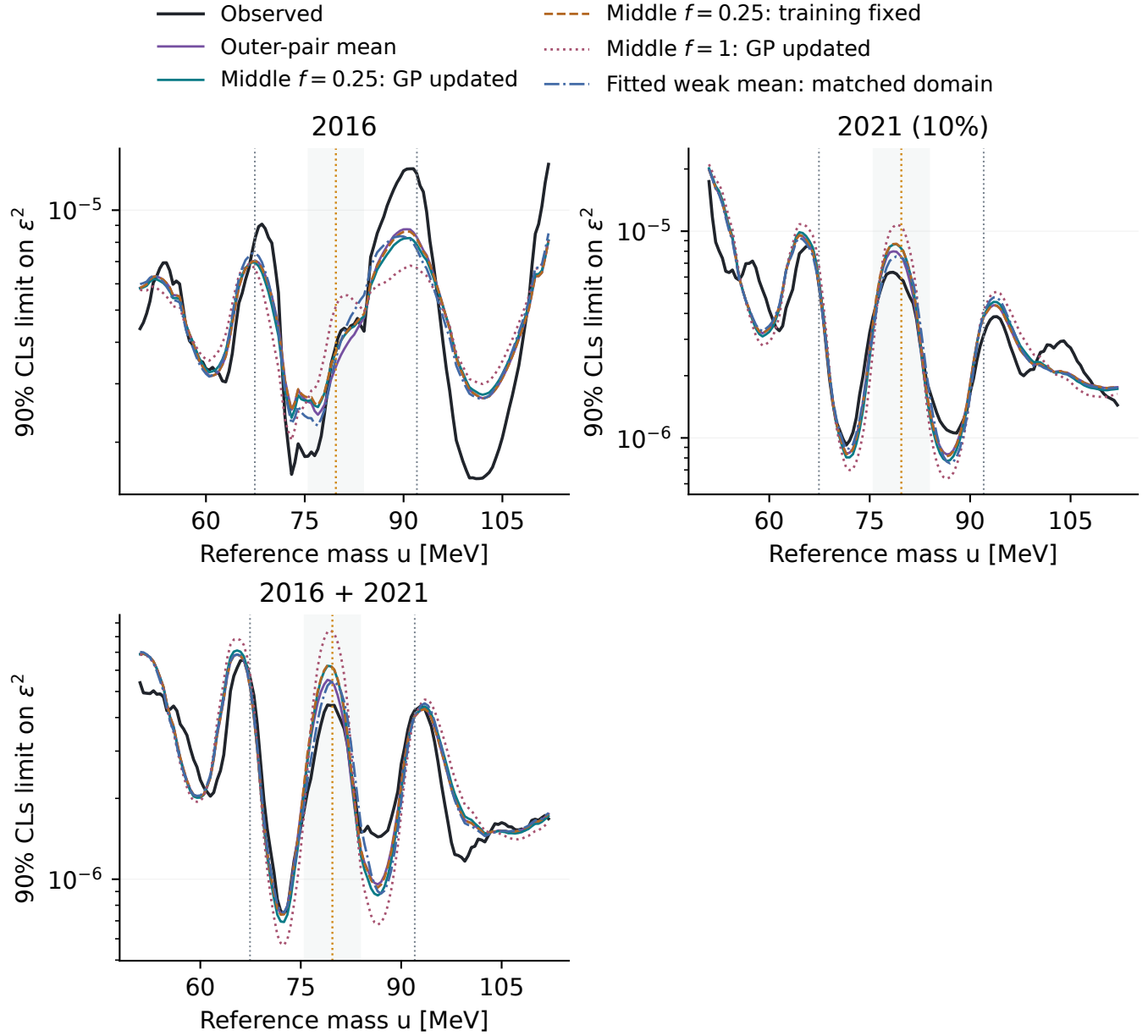
9. Lower triplet: 2016 + 2021 – spectra and component sizes



Complete count-spectrum construction. The benchmark middle signal has $f = 0.25$ at the parent midpoint, with unchanged outer peaks and continuum. All tails are retained. Curve labels distinguish injected benchmarks from refitted means; these objects answer different questions.

Campaign	m_M [MeV]	A_M	N_L	N_M	N_H
2016	79.72	77.09	9939	1882	5623
2021	79.21	77.09	12504	3469	14579

10. Lower triplet: 2016 + 2021 – upper-limit comparison

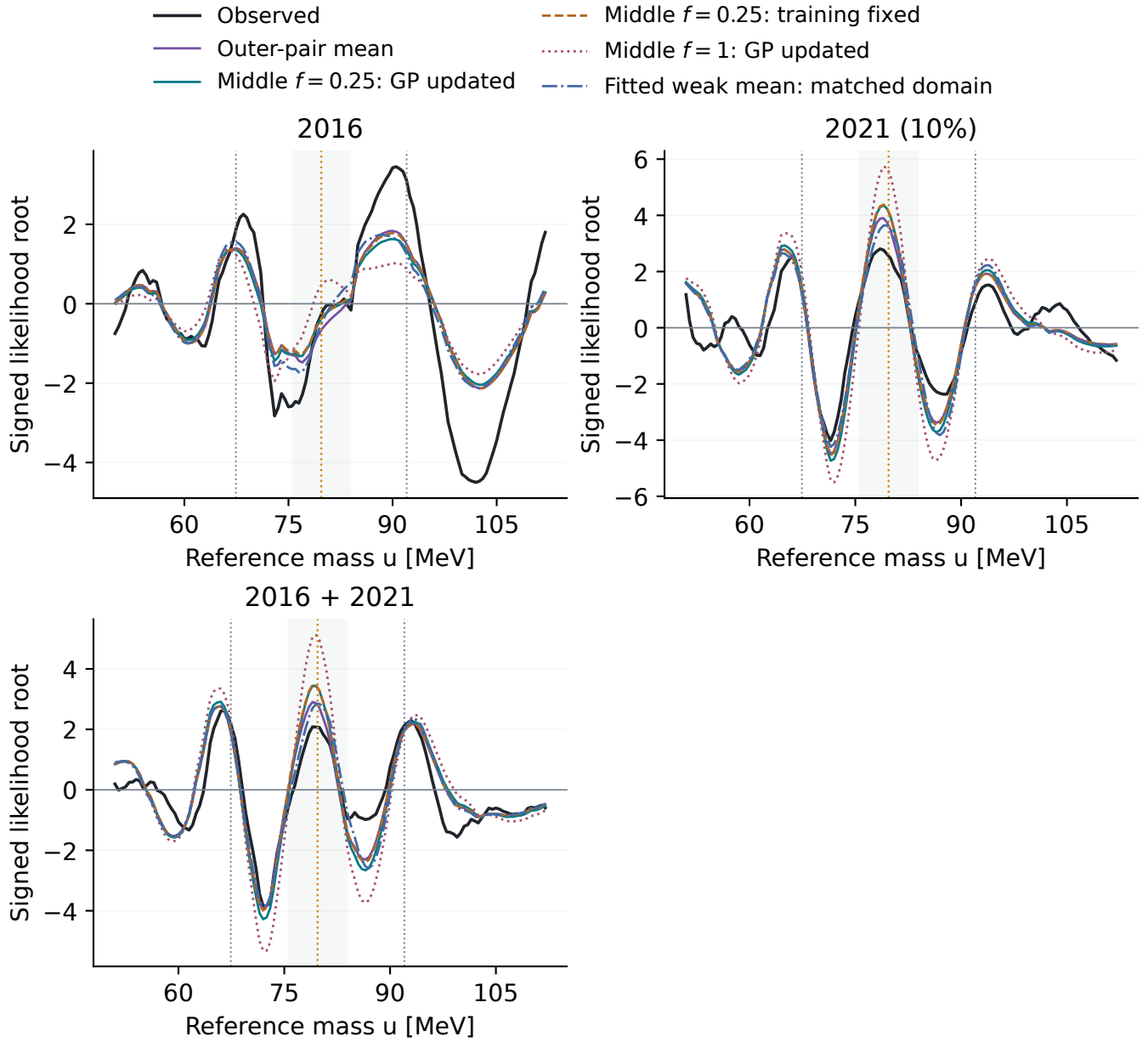


Actual 0.5 MeV bounded CL_s solutions for the observed spectrum, original pair and selected triplet constructions. The horizontal coordinate is the reference mass; each campaign uses its own shifted reconstructed test mass. Full fitted-mean controls refit the continuum and outer amplitudes, so their difference from the pair is not solely the added middle signal.

Figure 10 displays six plots showing the ratio of triplet to pair interaction strengths and positions for different reference masses u . The plots are arranged in a 3x2 grid. The rows represent different training sets: 2016, 2021 (10%), and 2016 + 2021. The columns represent the strength and position of the interaction. The x-axis for all plots is the reference mass u in MeV, ranging from 60 to 105. The y-axis for the left column is $U_{90}(\text{triplet})/U_{90}(\text{pair})$ and for the right column is $U_{90}(\text{triplet})/U_{90}(\text{pair})$. The legend indicates different values of f (0.1, 0.25, 0.5, 1) and u (75, 76, 77, 78, 79, 80) and a pair center of 79.72. The plots show oscillatory behavior with a central peak around $u = 79.72$ MeV.

12

12. Lower triplet: 2016 + 2021 – which trough is being filled?



The signed root exposes the initial sign of the response. In 2021 and the shared combinations, the parent pair already generates a positive central response. Adding a middle component can strengthen that predicted excess. Positive-signal p_0 would plateau at 0.5 for deficits and obscure this distinction; root curves retain it without implying a calibrated triplet significance.

13. Does filling the middle improve agreement with the observed curve?

The table compares the unmodified parent pair, the $f = 0.25$ midpoint injection, and the selected shared weak-triplet fitted mean. All entries are full-window RMS differences of logarithmic upper limits relative to the observed curve. They are descriptive distances, not independent-bin goodness-of-fit statistics. A local uplift can worsen the full curve or the neighboring troughs.

Reference	Search	Pair	Injected middle	Fitted weak triplet
All-three fit	2015	0.545	0.544	0.547
All-three fit	2016	0.326	0.342	0.317
All-three fit	2021 (10%)	0.250	0.279	0.256
All-three fit	All three	0.217	0.260	0.232
16+21 fit	2016	0.303	0.318	0.303
16+21 fit	2021 (10%)	0.257	0.286	0.257
16+21 fit	2016 + 2021	0.223	0.258	0.223

The matched-window shared weak fit gives a full-window combined distance of 0.232. The original pair gives 0.223; the fixed $f = 0.25$ midpoint injection gives 0.258. These reconstructions change different ingredients and cannot be ranked as particle probabilities.

For 2016 in the all-three construction, the benchmark moves the middle signed root from -0.723 to -0.429 , closer to the observed -0.365 , and raises the local limit by 11.1%. Yet its full-window discrepancy grows from 0.326 to 0.342. In 2021 the middle root instead grows from $+3.771$ to $+4.274$, farther from the observed $+2.614$. Thus the proposed mechanism works locally in 2016 while the same shared injection overfills the predicted 2021 central response.

A weak middle signal is therefore a plausible mechanism for softening an already negative middle response, but the count fit and whole-curve comparisons must determine whether that mechanism is useful for these data. The positive parent response in 2021 cannot be counted as a newly filled deficit. Likewise, the all-three shared fit's small improvement and boundary mass do not establish a resolved third resonance.

The parent continuum was itself fitted to these observed data, with substantial lower-family displacements from the external-GP continuum. Keeping it fixed makes the middle-signal intervention interpretable, but does not establish it as the true background. The separately refitted triplet means carry their own continuum and rate dependence. Event-level or independently constrained detector/background checks would be needed to distinguish particles from reconstruction and interpolation effects.

The scope of this bounded extension is intentionally local: outer masses and year shifts are pinned; the middle region is scanned on a declared grid; and the weak constraint is stated in ϵ^2 . This does not test every possible triplet or calibrate the many post-observation choices. No new toys, coverage claim or global significance are produced.

14. Validation, reproducibility and sources

The extension records 59,579 upper-limit solutions across 82 scenario scans, 96 primary and 44 matched-domain count-fit rows, 16 selected models and 38 full-support fitted spectra. All limits are finite, positive and numerically successful. The maximum absolute CL_s root residual is 1.59×10^{-8} . No random draws were made. Validation checks exact zero-middle identity, unchanged outer components/continuum, full-support yield normalization, coupling versus event-yield ledgers, aligned masks and templates, and paired frozen GP mean/covariance. Independent scalar likelihood replays compare saved limits with fresh solves. Count-fit checks verify nesting, the weak cone, positive means and the fixed parent GP experiment. Full-support exports retain zero amplitudes without floors or clipping. Numerical success does not calibrate the statistical interpretation.

The PDF is rendered page by page and checked for clipped text and figure legibility. A relocated source tree rebuilds the PDF and replays representative shifted limits. The source archive contains the pinned histograms, parent means, triplet fits, all response curves, metric tables, plotting code and checksum manifest. The parent PDF is preserved, and the combined study simply appends this extension to it.

The original 2016 support/qualification exception, fixed resolution and conversion, and exploratory 2015 extension to 100 MeV remain inherited. Frozen or interpolated kernel states do not imply frozen GP predictions. The previous ten-toys-per-scheme-and-strength pilot is unchanged; no sampling ensemble is added by this extension.

Sources

- [1] HPS GPR v5.2.2, *Shared separations, shifted peaks, and upper-limit troughs*, 10 September 2026, pinned as `inputs/parent/report.pdf`. This preserves the full numerical and interpretive context of the pair study.
- [2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, EPJC 71 (2011) 1554, [arXiv:1007.1727](https://arxiv.org/abs/1007.1727). The inherited solver profiles the count likelihood explicitly and uses its bounded asymptotic tails; no new triplet calibration is inferred.
- [3] Reproduction design and data: `PROTOCOL.md`, `README.md`, the numerical tables in `derived`, and the independent review records in `qa`.